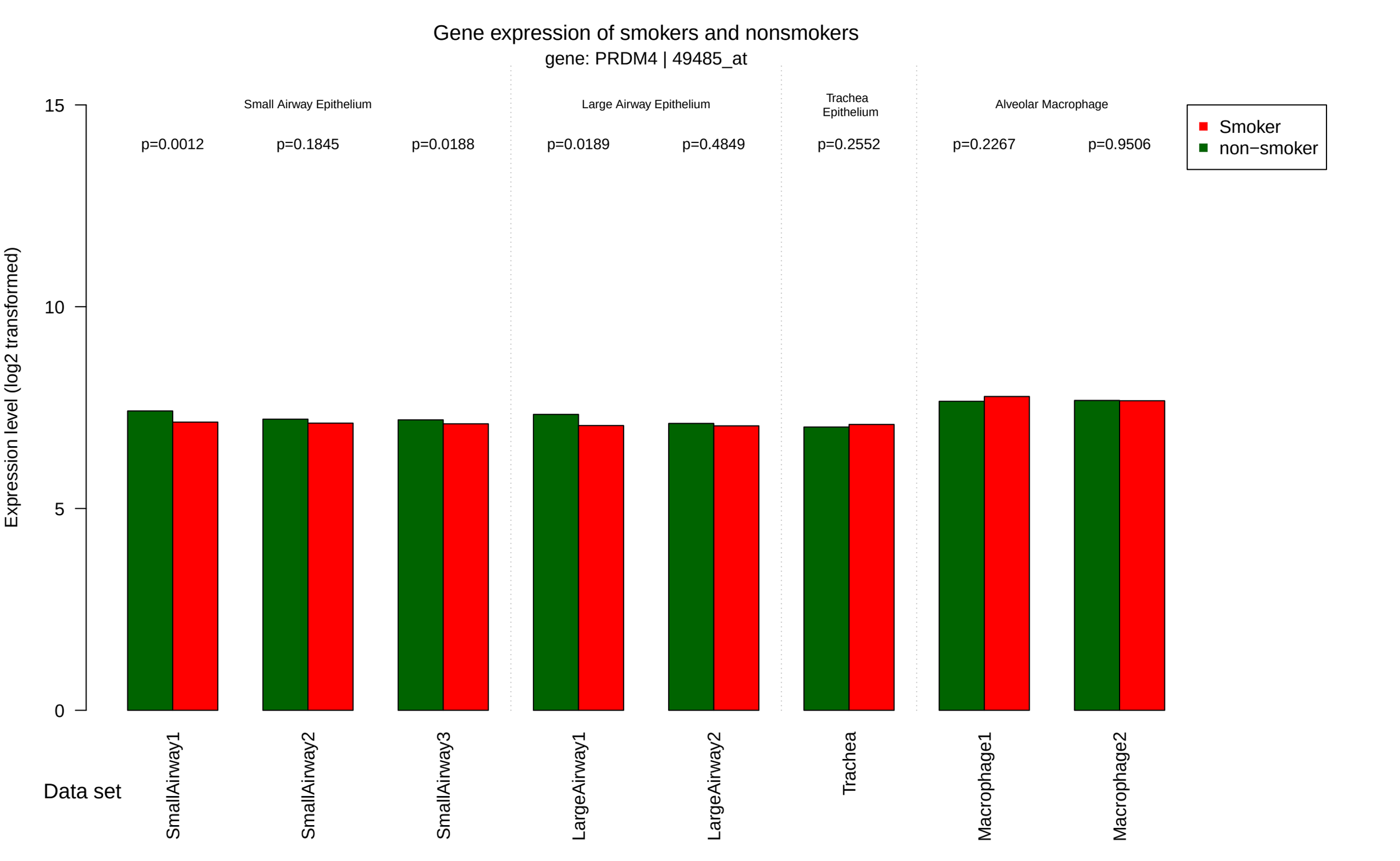
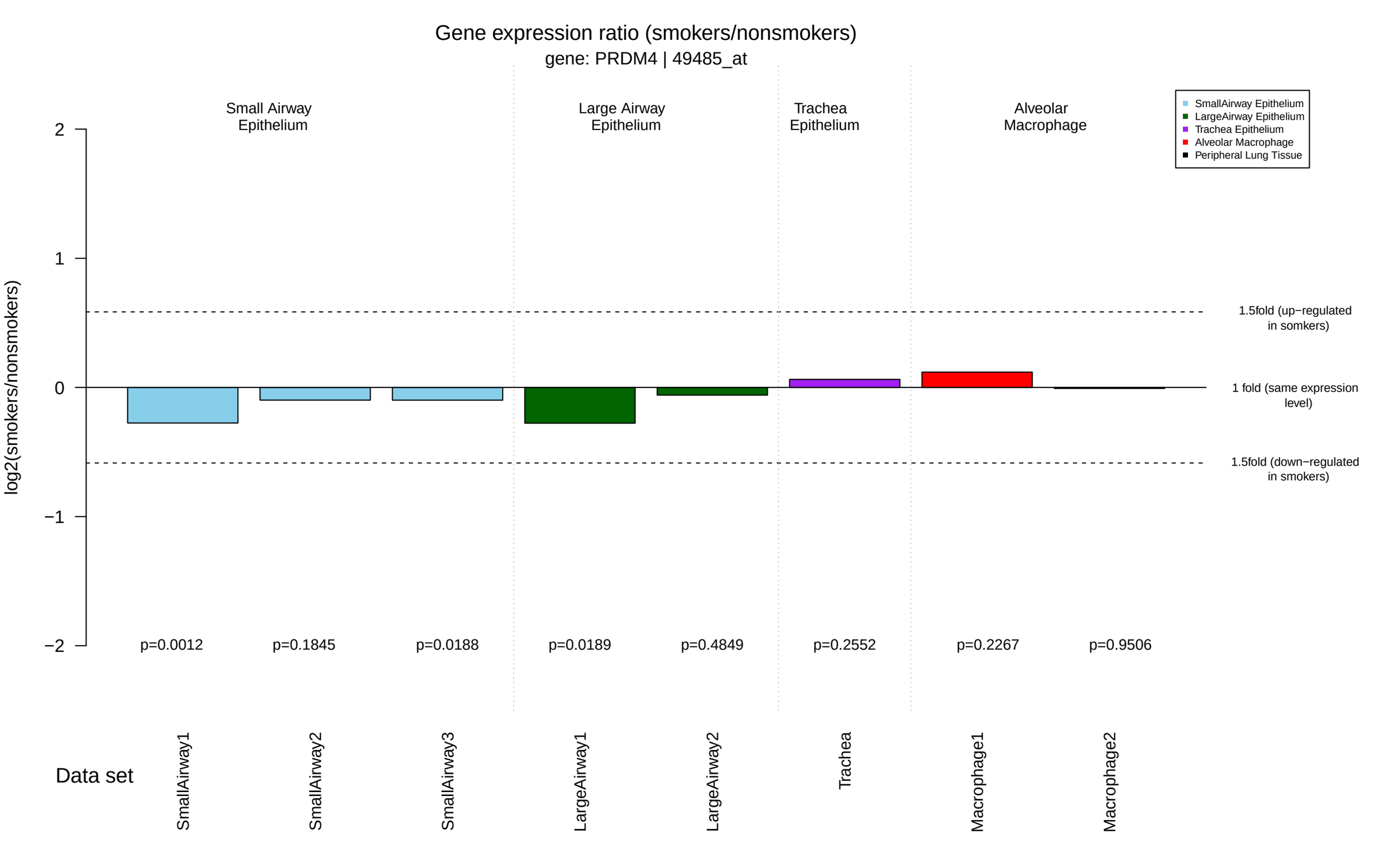
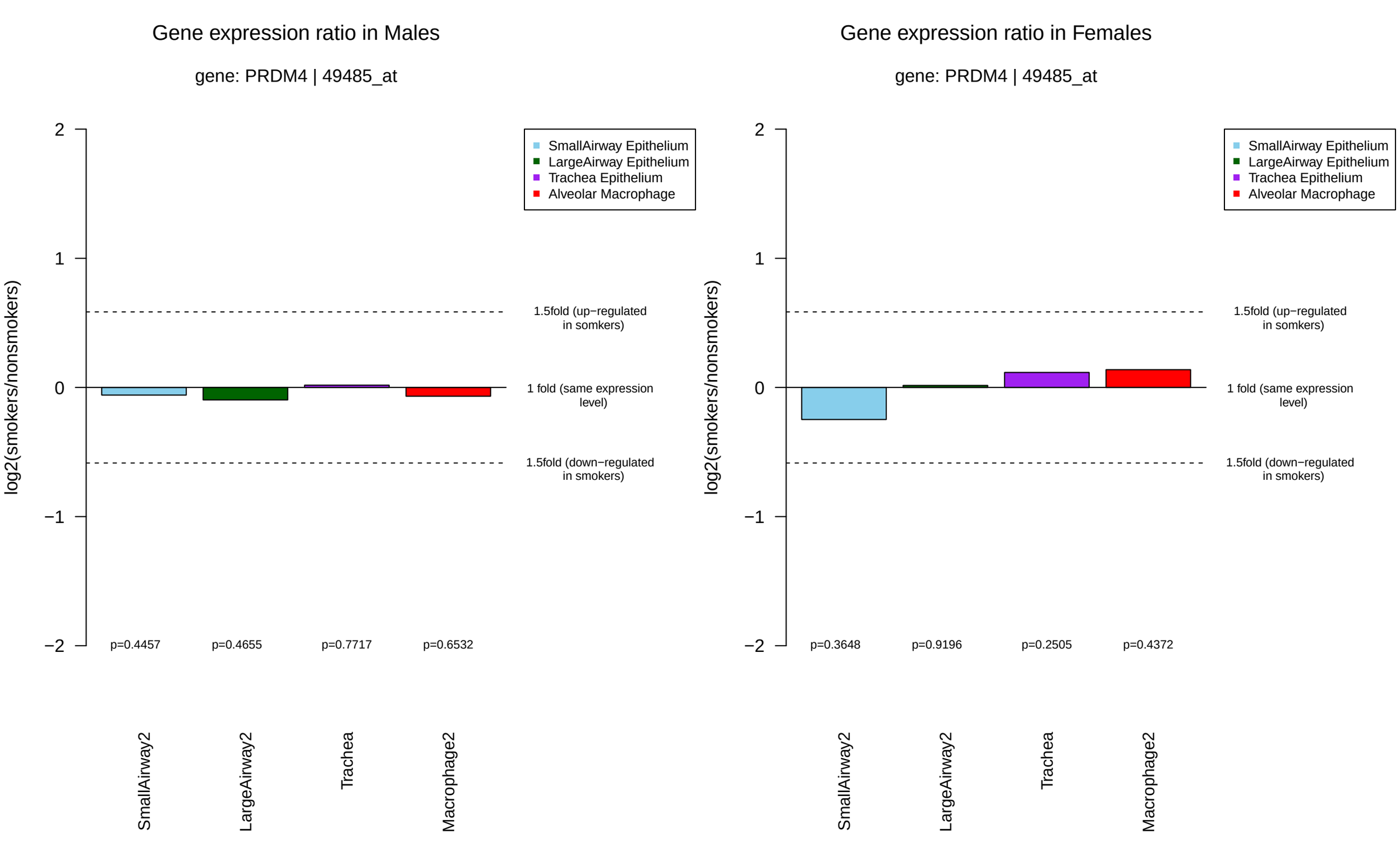
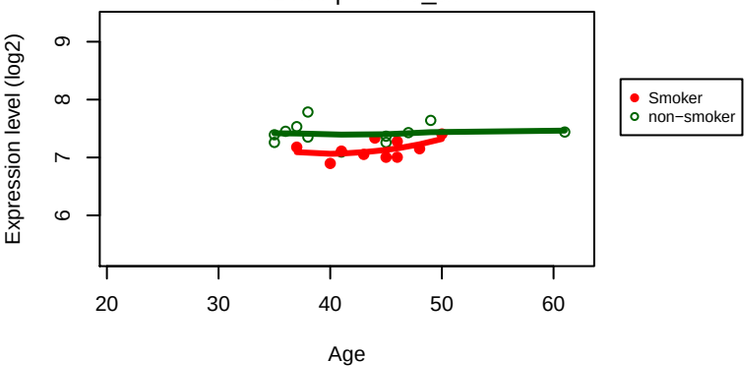
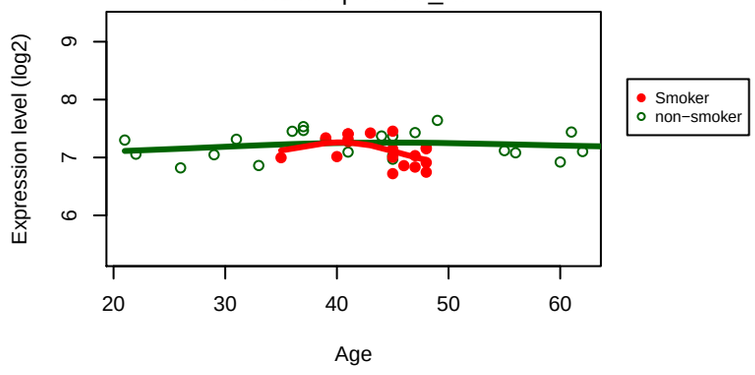
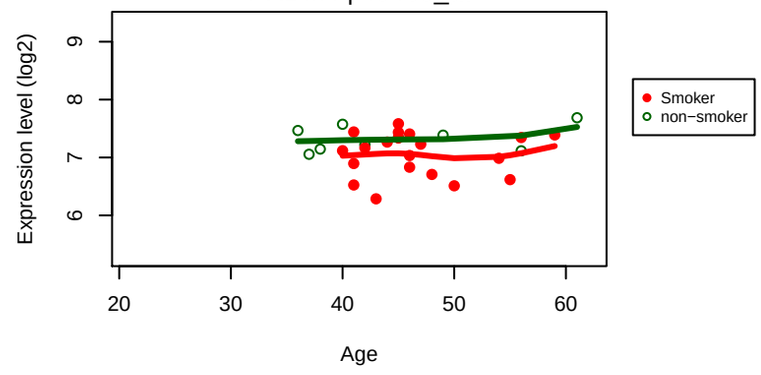

Gene expression of smokers and nonsmokers
gene: PRDM4 | 49485_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0012
p=0.1845
p=0.0188
p=0.0189
p=0.4849
p=0.2552
p=0.2267
p=0.9506
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

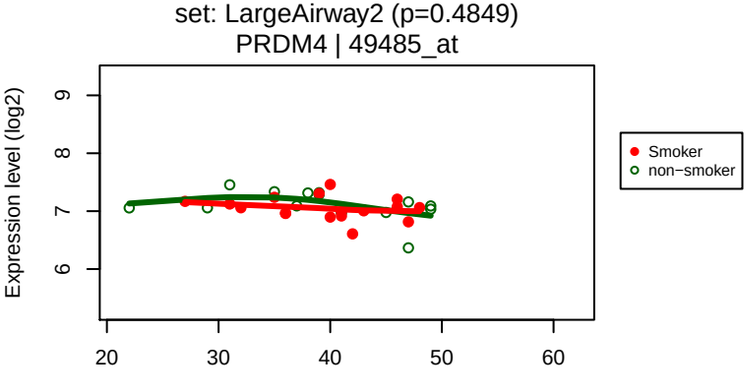

Gene expression ratio (smokers/nonsmokers)
gene: PRDM4 | 49485_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0012
p=0.1845
p=0.0188
p=0.0189
p=0.4849
p=0.2552
p=0.2267
p=0.9506
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

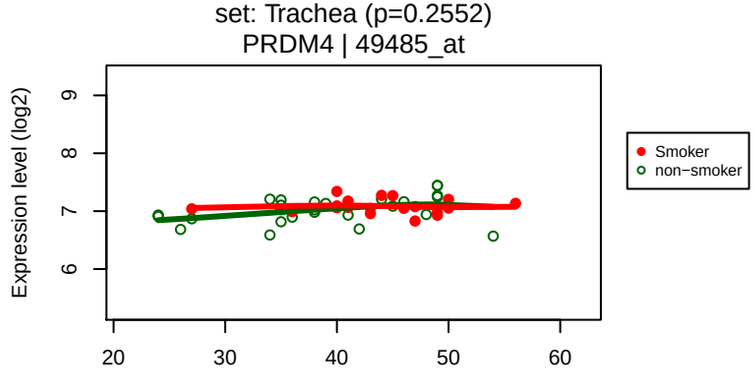

Gene expression ratio in Males
gene: PRDM4 | 49485_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4457
p=0.4655
p=0.7717
p=0.6532
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRDM4 | 49485_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3648
p=0.9196
p=0.2505
p=0.4372
SmallAirway2
LargeAirway2
Trachea
Macrophage2

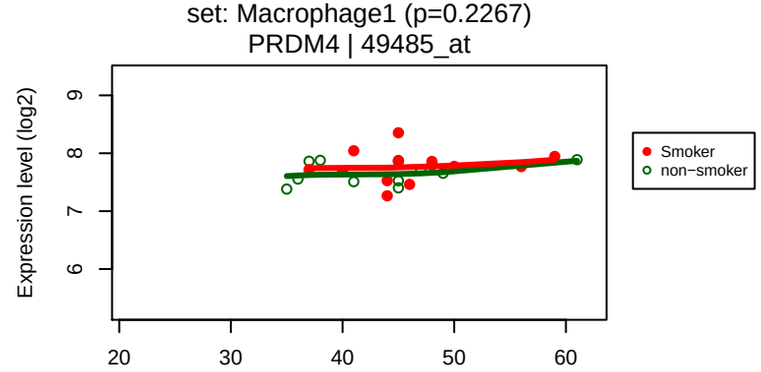

set: SmallAirway1 (p=0.0012)
PRDM4 | 49485_at
Expression level (log2)
Age
Smoker
non−smoker

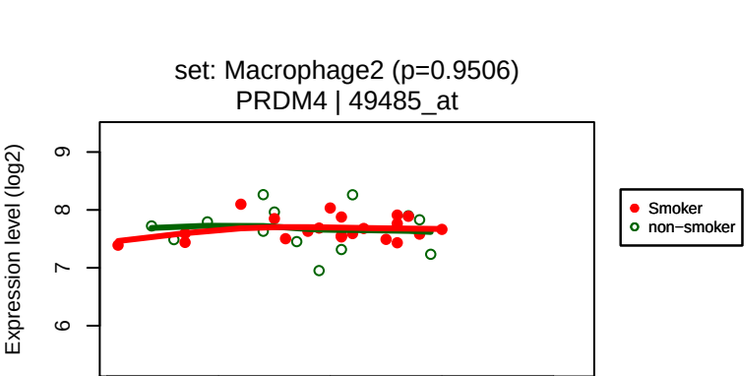

set: SmallAirway2 (p=0.1845)
PRDM4 | 49485_at
Expression level (log2)
Age
Smoker
non−smoker

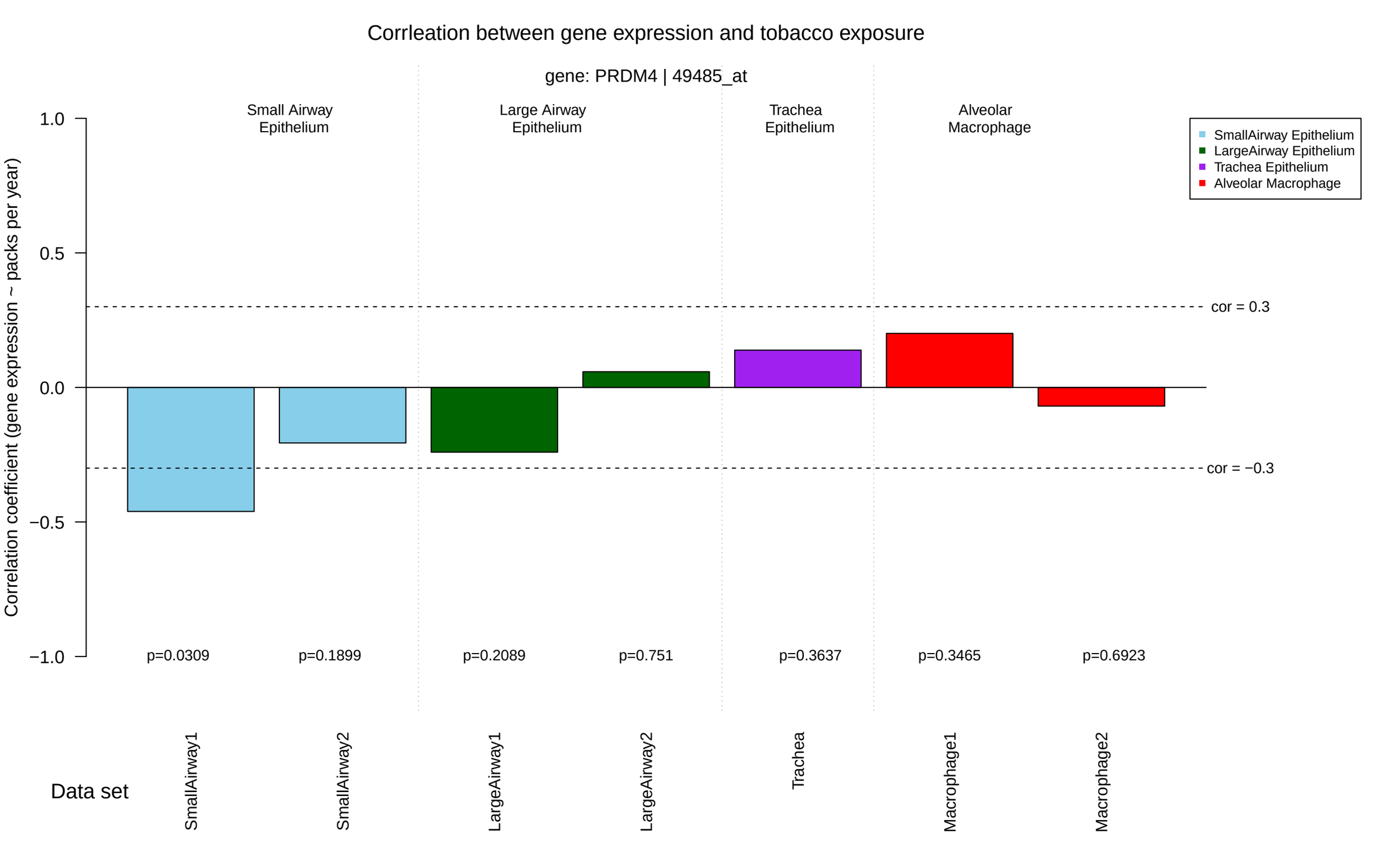

set: LargeAirway1 (p=0.0189)
PRDM4 | 49485_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4849)
PRDM4 | 49485_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2552)
PRDM4 | 49485_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2267)
PRDM4 | 49485_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9506)
PRDM4 | 49485_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRDM4 | 49485_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0309
p=0.1899
p=0.2089
p=0.751
p=0.3637
p=0.3465
p=0.6923
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2